DSA0502 – Query Processing for Data Science	
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LAB PROGRAMS 11 AND 12

Program 11

Question: Create a dataframe of ten rows and four columns with random values. Convert some values to NaN and highlight NaN values.

Aim: To implement the given Pandas program successfully.

Dataset :

Random DataFrame (10 x 4) generated using NumPy.

Code

```
import pandas as pd,numpy as np
df=pd.DataFrame(np.random.randn(10,4));df.iloc[1,2]=np.nan
df.style.highlight_null('yellow')
```

Output

NaN highlighted.

	0	1	2	3
0	-0.348404	-0.391820	1.421237	-1.081310
1	-0.205963	-0.580169	nan	0.924899
2	-0.030062	0.587078	-0.552198	-0.926955
3	0.506594	-0.971580	0.512773	0.031441
4	-0.423644	-3.080515	2.392626	-0.694340
5	1.660524	1.460418	-0.835879	-0.912986
6	0.468352	-0.548479	-0.512869	0.422191
7	1.008781	-1.495282	-0.784089	0.378129
8	1.503624	0.397760	0.178835	-0.521215
9	2.116600	-0.749646	-2.139734	-0.256790

Result

Thus, the program was executed successfully and the expected output was obtained.

Program 12

Question: Create a dataframe of ten rows and four columns with random values. Set dataframe background black and font yellow.

Aim: To implement the given Pandas program successfully.

Dataset :

Random DataFrame (10 x 4) generated using NumPy.

Code

```
import pandas as pd,numpy as np
df=pd.DataFrame(np.random.randn(10,4))
df.style.set_properties(**{'background-color':'black','color':'yellow'})
```

Output

Black background, yellow font.

	0	1	2	3
0	-0.982990	0.701889	-1.245160	1.523641
1	0.713842	0.360646	0.658665	-0.210780
2	0.693399	0.298173	0.396511	-0.598116
3	-1.380054	-0.301291	0.155390	0.816748
4	-1.721451	-1.352064	1.042921	0.180467
5	-0.738819	1.396515	-0.093485	0.267696
6	-0.687679	-0.500063	0.537522	0.686483
7	1.450588	-1.082532	1.782225	0.636566
8	-0.224290	-1.552851	1.105861	0.422853
9	2.064720	0.575817	0.395295	-0.067200

Result

Thus, the program was executed successfully and the expected output was obtained.